

Mathematics (041)

Reconstructed Question Paper

Note: This is a LaTeX reconstruction from the uploaded scanned solution booklet. Only question statements that could be read or reliably reconstructed have been included. Missing/obscured questions have not been invented.

SECTION B — Vectors / Calculus

Q.21 A vector $\vec{a}$ makes an angle θ with all three coordinate axes. If the magnitude of $\vec{a}$ is $5\sqrt{3}$, find the vector $\vec{a}$.

Q.22 Evaluate:

$$\int_0^{\pi/4} \sqrt{1 + \sin 2x} \, dx.$$

Q.23 Let

$$f(x) = \sin x - ax + b.$$

Find the range of values of a for which $f(x)$ is increasing on $\mathbb{R}$.

Q.24 Given two vectors $\vec{a}$ and $\vec{b}$ satisfying

$$\vec{a} = (x - 2)\vec{a} + \vec{b},$$

$$\vec{b} = (3 + 2x)\vec{a} - 2\vec{b},$$

find x , given that $\vec{a}$ and $\vec{b}$ are parallel/collinear.

SECTION C / D — Relations, Differential Equations, Probability and Integration

Q.27 Let R be a relation on $\mathbb{N}$ defined by mRn if and only if m is a multiple of n . Determine whether R is reflexive, symmetric and transitive.

Q.28 Solve the following linear programming problem graphically:

$$\text{Minimize } z = x - 5y$$

subject to

$$x, y \geq 0,$$

$$x - y \geq 0,$$

$$-x + 2y \geq 2,$$

$$x \geq 3,$$

$$y \leq 4.$$

Q.29 Given

$$x\sqrt{1+y} + y\sqrt{1+x} = 0,$$

find $\frac{dy}{dx}$.

Q.30 A die is thrown twice. Let X denote the number of times the number 2 appears. X can take the values 0, 1, or 2. Find the probability distribution of X and hence find $E(X)$.

Q.31 Evaluate:

$$\int \frac{1}{x} \sqrt{\frac{x+a}{x-a}} dx.$$

SECTION D / E — Integration, Matrices, Relations and Probability

Q.33 Evaluate:

$$\int \frac{x^2 + x + 1}{(x+2)(x^2+1)} dx.$$

Q.34 Find the shortest distance between the two lines:

$$\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$$

and

$$\frac{x-3}{2} = \frac{y+15}{-1} = \frac{z-9}{5}.$$

Q.35 Let

$$A = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}.$$

(a) Find AB and hence find B^{-1} .

(b) Represent the corresponding system of linear equations in matrix form and solve it.

Q.36 Let

$$S = \{S_1, S_2, S_3, S_4\}$$

and

$$T = \{T_1, T_2, T_3\}.$$

(i) Find the number of relations from S to T .

(ii) For

$$f = \{(S_1, T_1), (S_2, T_2), (S_3, T_2), (S_4, T_3)\},$$

determine whether f is one-one/injective and whether it is onto/surjective. Hence state whether f is bijective.

Q.37 Let E_1 , E_2 and E_3 denote the events that Amber, Bonzi and Comet manufacture the cars. Let A denote the event of selecting an electric car. Given

$$P(E_1) = 60\%, \quad P(E_2) = 30\%, \quad P(E_3) = 10\%,$$

$$P(A | E_1) = 20\%, \quad P(A | E_2) = 10\%, \quad P(A | E_3) = 5\%,$$

find:

- (i) the probability that a randomly selected car is an electric car;
- (ii) the probability that a randomly selected electric car was manufactured by Comet;
- (iii) the probability that a randomly selected electric car was manufactured by Amber or Bonzi.

Q.38 Let

$$f(x) = e^x \sin x$$

on $[0, \pi]$. Determine the intervals on which f is increasing/decreasing and identify the local maximum/minimum points.

Questions 1–20, 25–26, 32, and some graphical/subparts are not included because their original printed wording was not sufficiently recoverable from the uploaded solution scan.